

Intelligent Agents

Homework #3

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Show a case in which an IR contract net protocol only respectively based on O,C,S,M-type contracts will be stuck at a local optimality.

Say we have a set of agents $\mathcal{A}$ that only consists of 2 agents, let us refer to them as agent 1 and agent 2. And there are a total of 3 tasks, $T = \{t_1, t_2, t_3\}$. The initial allocation is $T_1 = \{t_1\}$ and $T_2 = \{t_2, t_3\}$. Let the task handling costs be $c_1(\{\emptyset\}) = 0$, $c_2(\{\emptyset\}) = 0$, $c_1(\{t_1\}) = 3$, $c_1(\{t_2\}) = 4$, $c_1(\{t_3\}) = 4$, $c_1(\{t_1, t_2\}) = 9$, $c_1(\{t_1, t_3\}) = 8$, $c_1(\{t_2, t_3\}) = 3$, $c_2(\{t_1\}) = 2$, $c_2(\{t_2\}) = 2$, $c_2(\{t_3\}) = 2$, $c_2(\{t_1, t_2\}) = 9$, $c_2(\{t_1, t_3\}) = 9$, $c_2(\{t_2, t_3\}) = 6$, $c_1(\{t_1, t_2, t_3\}) = \infty$, $c_2(\{t_1, t_2, t_3\}) = \infty$. We define increasing welfare as decreasing the sum of the costs paid by each agent in the agent set. In this case, the cost for both agents 1 and 2 from the initial allocation is 9 total. We can also see from the task handling costs for all task subsets that the only possible deal to increase welfare is where agent i gives t_1 to agent 2 and agent 2 gives $\{t_2, t_3\}$ to agent 1, creating a subsequent allocation of $T_1 = \{t_2, t_3\}$ and $T_2 = \{t_1\}$. However, this exchange is can be called something like an "asymmetrical swap" and can not be achieved by either one of O-, C-, S-, or M-contracts alone. It is not "one at a time" or "cluster" because there is a bilateral exchange going on (more than one task going both from agent 1 to 2 and from 2 to 1). Swaps describe this kind of bilateral exchange, but this is not a formally defined swap (S-contract) either, since swaps only deal with the swap of a single task (symmetrical exchange) but the welfare-increasing deal gives one task and gets two in return, and vice versa - so it cannot be considered a swap. Lastly, this step cannot be achieved by what is considered multiagent contracts either since multiagent contracts have to involve 3 or more agents and there are only 2 agents in this set. $\square$